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THE CAPACITATED r-ALLOCATION p-HUB MEDIAN PROBLEM

The p-hub median problem is a cornerstone of hub location theory, traditionally studied under two extreme allocation strategies: single allocation ($r = 1$) and multiple allocation ($r = p$). The r-allocation variant bridges this gap by allowing each non-hub node to connect to at most r hubs, offering a trade-off between network flexibility and operational simplicity. This essay formalises the **Capacitated r-Allocation p-Hub Median Problem** (CrApHMP), extending the uncapacitated model with capacity limits at hub facilities. We present a Mixed-Integer Linear Programming (MILP) formulation tailored for ease of implementation in Python's PuLP library, enabling experiments with parameter values and solution structures.

1 Introduction

Hub location problems arise in the design of many-to-many distribution systems, where flows between origins and destinations are consolidated at hub facilities to exploit economies of scale. The classic p-hub median problem selects p hubs from a set of candidate nodes and routes all traffic through them, minimising total transportation cost.

A critical design decision is the *allocation strategy*: how non-hub nodes connect to hubs. In the **single allocation** case ($r = 1$), each node is assigned to exactly one hub, simplifying operations but potentially increasing collection and distribution costs. In the **multiple allocation** case ($r = p$), each node may use any hub, maximising flexibility at the expense of consolidation and administrative complexity. The **r-allocation** problem, introduced by Yaman (2011), interpolates between these extremes: each non-hub node is allocated to *at most* r hubs, where $1 \leq r \leq p$.

The Capacitated r-Allocation p-Hub Median Problem captures a design trade-off that is ubiquitous in real-world logistics and infrastructure planning. In many-to-many distribution systems (e.g., express parcel delivery, air cargo, freight rail, and telecommunication networks), flows are consolidated at a small number of hub facilities to exploit economies of scale, but these facilities are rarely unconstrained: sorting centres have finite throughput, airports have limited slot and apron capacity, and network switches have bounded bandwidth. The r-allocation rule adds a second layer of realism by allowing each origin and destination to connect to more than one hub (improving resilience against disruptions and reducing collection and distribution costs) while still enforcing a limit r that prevents the operational and administrative complexity of fully flexible multiple allocation.

This makes the model particularly relevant for designing *resilient* and *cost-efficient* networks, where a single hub failure should not isolate a region, and for planning *hybrid* systems in which some regions justify redundant connections to several hubs while others do not. By tuning p, r, and the capacity parameters B_k, planners can explore the entire spectrum from lean, single-allocation designs to flexible, multi-hub architectures, making the CrApHMP a versatile tool for strategic network design under practical constraints.

While the uncapacitated r-allocation problem has received attention, practical applications often impose capacity limits on hubs. Cargo terminals, sorting facilities, and telecommunication switches have finite throughput capacities. This essay formalises the **Capacitated r-Allocation p-Hub Median Problem** (CrApHMP) and provides a MILP formulation.

2 Problem Statement

Consider a set of n nodes, $N = \{1, \ldots, n\}$, representing origin/destination points and candidate hub locations. We are given:

- $w_{ij} \geq 0$: flow demand from node i to node j (with $w_{ii} = 0$).
- $d_{ij} \geq 0$: distance (or unit transport cost) between nodes i and j.
- $\alpha \in (0, 1]$: inter-hub transfer discount factor (economies of scale).
- p: number of hubs to locate.
- r: maximum number of hubs each node may be allocated to ($1 \leq r \leq p$).

- B_k: capacity of hub k (maximum total flow that can enter the hub).

The cost of routing one unit of flow from i to j via hubs k and l is:

$$C_{ijkl} = \lambda\, d_{ik} + \alpha\, d_{kl} + \delta\, d_{lj}$$

where λ and δ are collection and distribution cost coefficients (often set to 1). If $k = l$, the flow passes through a single hub: $C_{ijkk} = d_{ik} + d_{kj}$ (assuming $\lambda = \delta = 1$ and no inter-hub discount).

The objective is to select p hubs, allocate each non-hub node to at most r hubs, route all flow through the hub network without violating hub capacities, and minimise total cost.

To help build intuition, consider a small instance with $n = 7$ nodes, $p = 2$ hubs, and $r = 2$. Figure 1 depicts a feasible (and in this case, optimal) solution. Nodes A and E are selected as hubs, shaded in dark grey. The remaining nodes are allocated to at most two hubs, shown by dashed grey edges (allocation links). The solid black arrows illustrate the routing of a representative flow from node B to node F: the flow is collected at hub A, transferred to hub E at the discounted rate α, and distributed to its destination F. Node C illustrates the flexibility of the r-allocation rule: it is connected to both hubs, providing redundancy, while node G is allocated only to E, reflecting a tighter, more consolidated design choice.

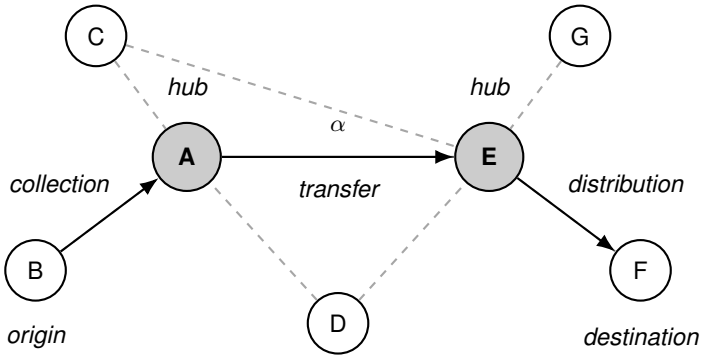


Figura 1: A feasible solution to a CrApHMP instance with $n = 7$, $p = 2$, $r = 2$. Hubs A and E (dark grey) serve all nodes. Dashed grey edges are allocation links; solid black arrows show a representative flow $B \to A \to E \to F$ with inter-hub discount α. Nodes C and D are allocated to both hubs ($r = 2$), while node B is allocated only to hub A, and nodes F, and G are allocated only to E.

3 MILP Formulation

We use the following decision variables:

$$z_{ik} = \begin{cases} 1 & \text{if node } i \text{ is allocated to hub } k \\ 0 & \text{otherwise} \end{cases} \qquad \forall i, k \in N \tag{1}$$

$$x_{ijkl} = \text{fraction of flow from } i \text{ to } j \text{ routed via hubs } k \text{ and } l \quad \forall i, j, k, l \in N \tag{2}$$

The formulation is:

$$(\mathcal{P}) \quad \min \quad \sum_{i\in N}\sum_{j\in N}\sum_{k\in N}\sum_{l\in N} w_{ij}\, C_{ijkl}\, x_{ijkl} \tag{3}$$

$$\text{s.t.} \quad \sum_{k\in N} z_{kk} = p \tag{4}$$

$$z_{ik} \le z_{kk} \qquad \forall i,k \in N \tag{5}$$

$$\sum_{k\in N} z_{ik} \le r \qquad \forall i \in N \tag{6}$$

$$\sum_{k\in N}\sum_{l\in N} x_{ijkl} = 1 \qquad \forall i,j \in N \tag{7}$$

$$\sum_{l\in N} x_{ijkl} = z_{ik} \qquad \forall i,j,k \in N \tag{8}$$

$$\sum_{k\in N} x_{ijkl} = z_{jl} \qquad \forall i,j,l \in N \tag{9}$$

$$\sum_{i\in N}\sum_{j\in N} w_{ij}\left(\sum_{l\in N} x_{ijkl} + \sum_{l\in N, l\neq k} x_{ijlk}\right) \le B_k z_{kk} \qquad \forall k \in N \tag{10}$$

$$z_{ik} \in \{0,1\} \qquad \forall i,k \in N \tag{11}$$

$$0 \le x_{ijkl} \le 1 \qquad \forall i,j,k,l \in N \tag{12}$$

Explanation of equations.

- (3): Minimise total transportation cost (objective function).
- (4): Exactly p hubs are opened ($z_{kk} = 1$ if node k is a hub).
- (5): A node can only be allocated to a node that is a hub.
- (6): Each node is allocated to at most r hubs. This is the defining constraint of the r-allocation model.
- (7): All flow from i to j must be routed through some pair of hubs (k, l).
- (8)–(9): Flow can only leave i through a hub k to which i is allocated, and can only enter j through a hub l to which j is allocated. These constraints link routing to allocation.
- (10): The total flow *entering* hub k (collection from all origins plus transfers from other hubs) cannot exceed capacity B_k. Note: the expression $\sum_{i,j} w_{ij} z_{ik}$ counts both outbound collection (first hub k) and inbound distribution (second hub k).

Collapsing the Formulation into a Concrete Instance The general MILP formulation $\mathcal{P}$ is deliberately stated over abstract sets N and indices i, j, k, l so that it applies to any instance. So, consider the *collapse* of $\mathcal{P}$ onto the specific graph of Figure 1. This makes the structure explicit and reveals which constraints are binding and which become redundant.

Let the node set be $N = \{A, B, C, D, E, F, G\},\ n = 7$, with the parameters $p = 2$, $r = 2$, inter-hub discount α, and capacities B_A, B_E. Flows w_{ij} and distances d_{ij} are instance data, with $w_{ii} = 0$ and $d_{ii} = 0$.

Decision variables. The binary allocation variables reduce to a 7×7 matrix: $z_{ik} \in \{0,1\},\ i,k \in \{A, B, C, D, E, F, G\}$. The routing variables x_{ijkl} form a $7^4 = 2401$-dimensional family, but in practice most are fixed to zero by the constraints.

4 Experimental Guidance

Given that, you should implement the given formulation using PuLP and use it to solve the given instances. Also, you should also explore the results (by cost) when varying r and p within reasonable bounds (see below). Finally, you should present your code and results in class.

1. **Effect of r.** Set $r = 1$ (single allocation), $r = p$ (multiple allocation), and intermediate values. Observe how total cost and allocation patterns change.
2. **Capacity tightness.** Vary B_k from loose to tight. Identify when the capacity constraints become binding and how hubs are redistributed.
3. **Discount factor.** Change α and observe the trade-off between hub consolidation and routing efficiency.
4. **Solution structure.** Print the allocation matrix z and the routing fractions x to visualise flow paths. For small instances, you can typically use `.dot` graph format and visualise it in `edotor.net`.

Regarding the instances, you will find the following testing sets:

The CAB (Civil Aviation Board) data set It is based on airline passenger flows between some important cities in the United States. It consists of a data file, presented by O'Kelly in 1987, with the distances and flows of a 25 nodes graph.

The AP (Australian Post) data set It is based on real data from the Australian postal service and was presented by Ernst and Krishnamoorthy in 1996. The size of the original data file is 200 nodes. Smaller instances are presented, though. Regarding the flows between nodes, these instances do not have symmetric flows (i.e., for a given pair of nodes i and j, w_{ij} is not necessarily equal to w_{ji}). Moreover, flows from one node to itself can be positive (i.e., w_{ii} can be strictly positive for a given i).

The USA423 data set This is a family of instances introduced based on real airline data. It consists of a data file concerning 423 cities in the United States, where real distances and passenger flows for an accumulated 3 months period are considered.

5 Conclusion

The Capacitated r-Allocation p-Hub Median Problem offers a rich yet tractable setting for teaching hub location theory. The MILP formulation presented here balances completeness with implementability: the four-index x variables are conceptually simple, and PuLP's declarative syntax handles the complexity. By varying p, r, and capacities, students can develop intuition for the fundamental trade-offs in hub network design.

Keep going. The ability to model a real problem as a set of variables and constraints is a superpower that few people develop, and you are developing it right now. Every hour spent wrestling with these ideas compounds into a way of thinking that will serve you far beyond this course. 🏠🧱🔍